

Counterfactual Explanations

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Our approach

- **From:** small sets of movies
- **To:** large sets sharing one clear property (genre / decade / agreement)
- We try to avoid iterative search for small sets and rather focus on different kinds of explanation

```

class Candidates(ABC):
    def __init__(self):
        super().__init__()

    @abstractmethod
    def __call__(self, table: pd.DataFrame) -> pd.Index:
        """Return a pandas Index (or list-like) of candidate item ids."""
        pass

    @abstractmethod
    def get_explanation_string(self, movie_id: int) -> str:
        """Return a human-readable explanation string for the given movie."""
        pass

```

We create different kinds of explanation based on the way we discover the **set of candidates being removed**

Decade candidates

- Using the titles, we extract years
- We group the movies by decades
- For candidates we select the decades users have most interacted (with normalization)

Genres candidates

- We group the movies by genres
- For candidates we select the genres users have most interacted (with normalization)

Most agreed items

- We select items that all users interacted with
- we select movies that vary in the rating the least

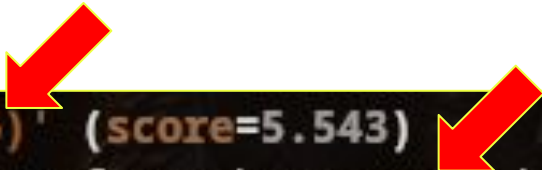
Pipeline

- For each removed candidate list **recompute** the recommendation
- See what **changed** in the final recommended list
- For each movie that **disappeared** provide explanation linked to the set of candidates
- Result will return multiple (or none) explanations for each movie, depending how much the recommendation was influenced by the list of candidates

Results & findings

Interesting behaviour for decades:

- We have decided to solve that by guardrails



```
[8236] 'While the City Sleeps (1956)' (score=5.543)  
1) If your group had not rated movies from the 2000s, then
```

Different behaviour depending of setting NaN and 0

- Could be used as additional explanation
 - **0** = Users disliked, **NaN** = Users have not rated

Results & findings

9. [5328] 'Rain (2001)' (score=5.333)

- 1) If your group had not rated movies from the 2000s, then movie 'Rain (2001)' would not be recommended.
- 2) If your group did not agree so strongly on movies like 'Matrix Reloaded, The (2003)', 'Lord of the Rings: The Fellowship of the Ring, The (2001)', 'Men in Black (a.k.a. MIB) (1997)', and others, then movie 'Rain (2001)' would not be recommended.
- 3) If your group had not watched any Comedy movies, then movie 'Rain (2001)' would not be recommended.

10. [3951] 'Two Family House (2000)' (score=5.333)

- 1) If your group had not rated movies from the 2000s, then movie 'Two Family House (2000)' would not be recommended.
- 2) If your group had not watched any Comedy movies, then movie 'Two Family House (2000)' would not be recommended.

Our results provide clear and intuitive explanations, but their quality relies on how well the Candidates method is implemented